

MARCUS CHITURA

24 BICCARD STREET, BRAAMFRONTEIN | Cell: 069 528 3950 | Email: chituramarcus@gmail.com

Projects Portfolio - <https://marcuschitura.github.io/Projects-Portfolio/>

PROFESSIONAL SUMMARY

A graduate Electrical Engineer with experience in designing and implementing engineering solutions through circuit design, PLC automation platforms and classic control systems. Comfortable developing Ladder Logic using Siemens TIA Portal (S7-1200) and Allen-Bredley RSLogix 500 integrating sensors, actuators and control hardware into automated systems. Bringing in a solid foundation in computational modelling and embedded design using tools like MATLAB/Simulink, LTSpice, Arduino IDE and C++ programming to support analysis and implementation. Driven by a passion for building smart, reliable and efficient engineering problem solving as showcased by my project's portfolio.

WORK EXPERIENCE

March 2024 – April 2024

Research Assistant | University of Witwatersrand

- Developed, validated and optimised FEMM models to simulate realistic magnetic behavior.
- Performed comparative analysis of multiple circuit configurations and documented findings in a technical report.

November 2023 – February 2024

Engineering Intern | CGM Consulting Engineers

- Provided electrical building services for three major projects that involved lighting design, solar system installations and electrical reticulation.
- Designed room lighting layouts using Dialux and AutoCAD for the Great Zimbabwe project, ensuring they met client and SANS 10114 regulations.
- Reviewed and corrected Bills of Quantities and prepared reports for the Great Zimbabwe project.
- Completed onsite inspections for the PUMA service station, while shadowing a senior engineer ensuring installed electrical services matched design intent and safety requirements.

EDUCATION

BSc Electrical Engineering | University of the Witwatersrand

- Relevant Coursework: Electrical Design I & II, Control Systems I & II, Signals & Systems I & II, Software Development I & II, Power Engineering, Data Structures & Algorithms, Electromechanical Conversion

TECHNICAL SKILLS

- Programming: C++, Python, MATLAB, R, Arduino IDE, Git, ARM Assembly Language, NS-3
- Simulation & Design: FEMM, MotorSolve, LTSpice, Proteus Circuit Simulator, Simulink & MATLAB, Siemens TIA Portal, Allen-Bredley RSLogix 500, Factory IO
- Electrical & CAD Tools: AutoCAD, Dialux Lighting Design, Proteus PCB Design
- General: Microsoft Office Suite, Quantitative Analysis, Technical Reporting, Problem Solving